From Regional Dialects to Standard Written Marathi: A Neural Sequence-to-Sequence Normalization Framework

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Abstract

Regional sub-dialects of Indic languages face severe shortage of data and limited results for text normalization. We investigate dialect normalization for Marathi texts by converting non-standard text from three regional dialects into Standard Written Marathi. We introduce an automated synthetic parallel data expansion pipeline that combines LLM-based translation with a multi-tier verification engine. We fine-tune mT5-small and IndicBART using a 5-fold cross-validation and evaluate their performance under varied configurations. Results show that the expanded multi-dialect dataset improves model performance, with mT5-small achieving 69.51 BLEU and 84.58 chrF++, that surpasses IndicBART by 12.39 BLEU. These findings demonstrate the effectiveness of verified synthetic data augmentation for improving Marathi dialect normalization in low-resource settings.

Keywords: Dialect Normalization, Low-Resource NLP, Marathi Dialects, Sequence-to-Sequence Modelling, Synthetic Data Augmentation, mT5, IndicBART.

1 Introduction

Rapid development of deep learning architectures and self-attention mechanisms (Vaswani et al., 2017) has made it possible to develop systems capable of understanding and generating natural language. Open-weight models have gained popularity to process text in various domains as they provide a finer control over system architecture, runtime environment, and training data (Dubey et al., 2024). Yet current language systems tend to work efficiently with standard high-resource languages. Queries based on non-standard input yield unexpected and undesired outputs (Joshi et al., 2020). Speakers of regional dialects face many issues in systems that use these models across automated speech recognition, machine translation and other language processing systems (Koenecke et al., 2020; Zampieri et al., 2020).

Dialect normalization is the task of translating non-standard regional dialects into standard written language. The main challenge to be addressed while designing such systems is the preservation of the meaning of original text (Blaschke et al., 2024). It serves to be an essential layer for digital linguistic inclusion. Countries having large and diverse speech communities, such as India, which has 22 official languages and hundreds of regional dialects spoken by 1.4 billion people, regional varieties have remained unattended in computational linguistic research. An example of such linguistic disparity is Marathi, an Indo-Aryan language spoken by over 83 million people in western India (Kulkarni-Joshi, 2023). Development of reliable pipelines for regional Indic dialects have faced these fundamental challenges:

1. **Resource Scarcity**: Parallel corpora mapping the regional dialects to standard written language do not exist. Curation of such sets by experts fluent in multiple dialects is both expensive and unscalable (Blaschke et al., 2024; Dhasmana et al., 2026).
2. **Morphological and Dialectal Heterogeneity**: Marathi exhibits rich morphology where different words undergo varied complex transformations across different regions where the dialects are spoken (Ansary et al., 2024). Consequently, evaluations vary significantly

across individual dialects—ranging from systematic suffix alternation in Varhadi to intense lexical and phonetic divergence in Northern and Southern Konkan (Joshi et al., 2025).
3. **Underperformance of Multilingual Models**: Performance of pre-trained models like IndicBART (Dabre et al., 2022), IndicNLP Suite (Kakwani et al., 2020), and mT5 (Xue et al., 2021) seems to drop drastically on non-standard language variants due to fragmentation during tokenisation and out-of-vocabulary phonological shifts (Joshi et al., 2020; Kumar et al., 2024).

Our work mainly tries to investigate and conclude (a) how do normalisation scores vary across regional Marathi dialects which have different degrees of variations as compared to standard Marathi; (b) how effectively can existing sequence-to-sequence architectures adapt to dialect-to-standard mapping under task-specific conditioning; and (c) whether automated synthetic data expansion can accurately solve the problem of data scarcity and eliminate cross-dialect hallucinations and grammatical errors (de Gibert et al., 2024; Ji et al., 2023). To address these questions, we construct an end-to-end normalization framework combining LLM-based synthetic generation, multi-tier syntactic and semantic verification, and conditioned transformer fine-tuning across both single-dialect and multi-dialect settings.

Our core contributions are:

- **Parallel Corpus Generation**: Based on transcripts from the RESPIN dataset (Kumar et al., 2025), we construct a parallel corpus for three of the Marathi dialects (Southern Konkan, Northern Konkan, and Varhadi) mapped to Standard Marathi.
- **Automated Synthetic Augmentation with Verification**: We design a data expansion pipeline paired with a multi-tier verification engine, which has rule-based script filtration and LLM-based semantic inspection.

2 Related Work

2.1 Regional Dialect Standardization across Language Families

Early approaches in the context of European languages used Character-Level Statistical Machine Translation (CSMT) and phrase-based models for Swiss German Dialects (Scherrer and Ljubešić, 2016) which were later replaced by pre-trained language models (Lusetti et al., 2018; Kresic and Abbas, 2024). Similar approaches have been developed for Finnish, Swedish, and Estonian dialects (Partanen et al., 2019; Hämäläinen et al., 2020, 2022). A study of four language families by Kuparinen et al. (Kuparinen et al., 2023) suggested that fine-tuned sequence-to-sequence transformers surpassed the rule-based and SMT baselines. Other than European countries, East Asian and African-Asian varieties have also shown similar patterns. Japanese regional dialects have been processed using phrase-level LSTM modules (Abe et al., 2018), while Mandarin scripts relied on acoustic-phonetic mapping (Zhao and Chodroff, 2022). The Conventional Orthography for Dialectal Arabic (CODA) framework (Habash et al., 2012) in the landscape of Arabic languages gave guidelines for Modern Standard Arabic (MSA) which led to multi-city translation benchmarks (MADAR (Bouamor et al., 2018), NADI (Abdul-Mageed et al., 2023)) and morphology-oriented prompt design for LLMs (Eryani et al., 2020; Dimakis et al., 2025).

2.2 Indic Dialect Landscape and Marathi NLP

Even though pre-trained models such as IndicBART (Dabre et al., 2022) and mT5 (Xue et al., 2021), which are trained on multilingual corpora cover major languages, they lack understanding of regional sub-dialects. Models based on a single language like L3Cube-MahaNLP (Joshi et al., 2022) offer pre-trained Marathi models for classification, but do not support generative text standardization. INDIC-DIALECT (Kumar et al., 2024) considers Hindi varieties (Bhojpuri, Magahi, Maithili), for evaluation via WFST frameworks for text normalization while preserving the underlying semantics (Pratap et al., 2024). However, Marathi dialects were primarily dealt for acoustic/ASR classification (Mulla and Kumar, 2022; Pawar et al., 2026), or zero-shot LLM retrieval (Gaikwad et al., 2025; Vyavhare et al., 2026). Our work fills this gap by presenting an end-to-end neural fine-tuning pipeline with evaluations and data augmentation strategies.

2.3 LLM Distillation and Multi-Tier Verification

LLMs can be used to suffice for data scarcity in various tasks (Wang et al., 2022; Taori et al., 2023; Mukherjee et al., 2023). Knowledge distillation strategies can be employed to train compact and efficient models based on the outputs generated by a larger teacher model (Hsieh et al., 2023; Pecher et al., 2024; de Gibert et al., 2024). But the use of such techniques for low-resource regional dialects involves significant amount of hallucinations and noise (de Gibert et al., 2024; Ji et al., 2023). We solve this by utilising Gemma-4 (Google DeepMind Gemma Team, 2024) with a robust morphologically aware system prompt and a multi-tier verification system that ensures integrity of the data generated.

3 Methodology

3.1 Original Corpus Curation and Foundation in RESPIN

Base sentences for our task are drawn from the domain-specific transcripts of the RESPIN-S1.0 initiative (Kumar et al., 2025), developed at IISc Bangalore under the Gates Foundation project. While RESPIN has audio recordings and their transcripts across agriculture and banking domains for Indian languages and their dialects, it lacks aligned parallel dialect-to-target language translations required for efficient normalisation. The large speaker diversity ensures broad coverage of intra-dialect phonetic and lexical variation, while the balanced gender and domain splits reduce systematic bias in our derived parallel corpus. Detailed corpus statistics and demographic splits are summarized in Table 1 and Figure 1.

Table 1: RESPIN-S1.0 Marathi Source Corpus Statistics

Code	Dialect Name	Utterances (%)	Duration (h)	Speakers	Unique Prompts
D1	Southern Konkan (Malvani)	203,651 (25.14%)	247.31	732	5,838
D2	Northern Konkan (Ahirani)	198,560 (24.52%)	265.25	557	5,825
D3	Standard Marathi (Puneri)	208,850 (25.79%)	255.79	608	6,077
D4	Varhadi (Vidarbha)	198,873 (24.55%)	257.70	747	5,330
Total	**All Dialects**	**809,934 (100%)**	**1,026.06**	**2,644**	**23,069**

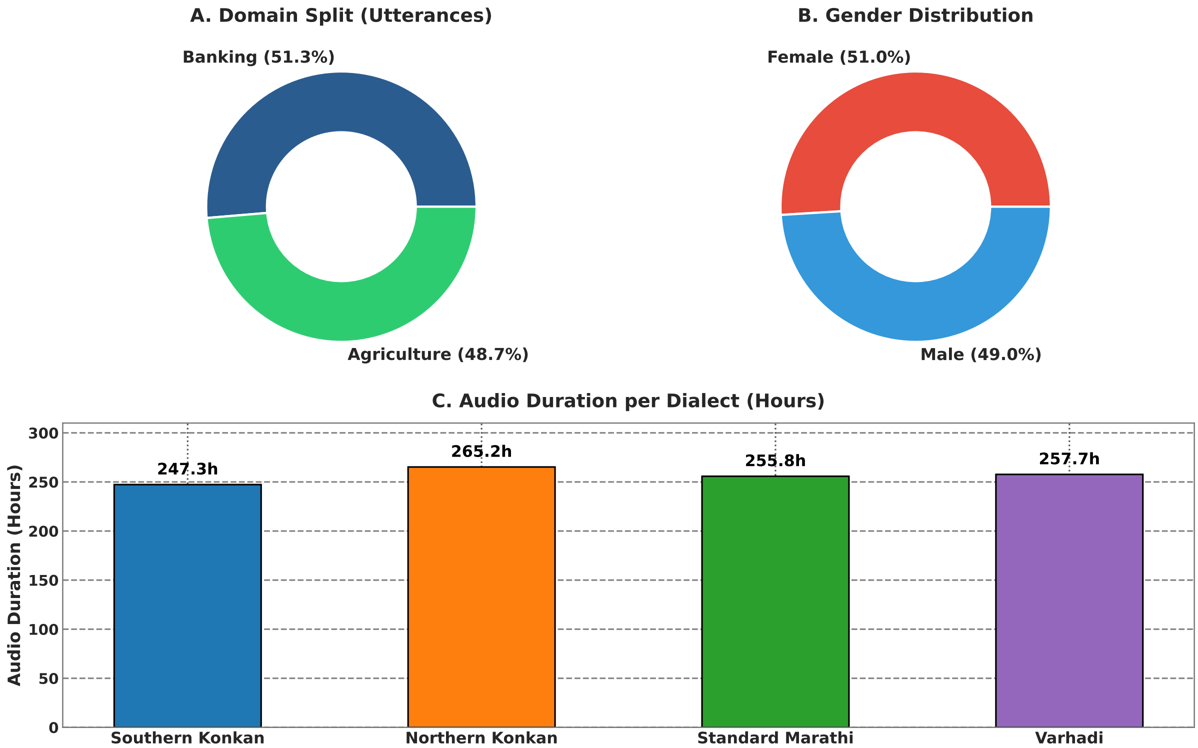

Figure 1: RESPIN-S1.0 Marathi source corpus demographics: (A) domain distribution; (B) gender split; (C) audio recording duration per dialect variant.

3.2 Target Dialects and Linguistic Properties

We target the following major Marathi dialects that span the primary regions of Maharashtra and exhibit distinct linguistic and phonetic patterns based on the RESPIN corpus:

1. **Southern Konkan (Konkan Coast)**: Characterized by shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms. Lexical differences include systematic vowel shifts: standard जातो ("he goes") becomes Southern Konkan जातां; standard कुठे ("where") becomes कुठंय. Domain-specific terminology generally remains untouched but often accompanied by Konkani words.
2. **Northern Konkan (Khandesh Region)**: Heavily influenced by neighbouring Gujarati and Bhili language families, thus it exhibits significant consonant simplification and dental consonant shifts. Notable transformations include: standard कशाला ("why") → Northern Konkan कसाले; standard मुलगा ("boy") → Northern Konkan पोर. Northern Konkan possesses the highest degree of lexical divergence among the three target dialects.
3. **Varhadi (Vidarbha/Eastern Maharashtra)**: Spoken in Vidarbha, it exhibits characteristic differences and verb suffix modifications. Standard interrogative का? ("why?") becomes Varhadi काऊन?; verbal forms undergo regular suffix mapping patterns. Among the three dialects, Varhadi maintains closest structural similarity to Standard Marathi as transformations are more systematic.

3.3 LLM-Driven Translation and Verification Pipeline

To deal with data scarcity, we developed an automated synthetic data translation and generation pipeline powered by Gemma-4 (Google DeepMind Gemma Team, 2024) with 12B parameters configuration, accessed via Ollama. The complete end-to-end normalization architecture is illustrated in Figure 2. We considered several design decisions to ensure generation quality to be maximised:

- **1-Call-Per-Sentence Architecture**: Instead of generating batches of parallel pairs per API call which could risk contamination of context, each dialectal sentence is processed separately with a self-contained prompt.

- **Detailed Prompting Strategy**: The generation prompt enforces semantic fidelity rules: the standard output must preserve exact meaning, domain terminology (agriculture/banking), named entities, and numerical values from the input. The prompt includes dialect-specific lexical mapping tables and 5-8 few-shot examples per dialect.

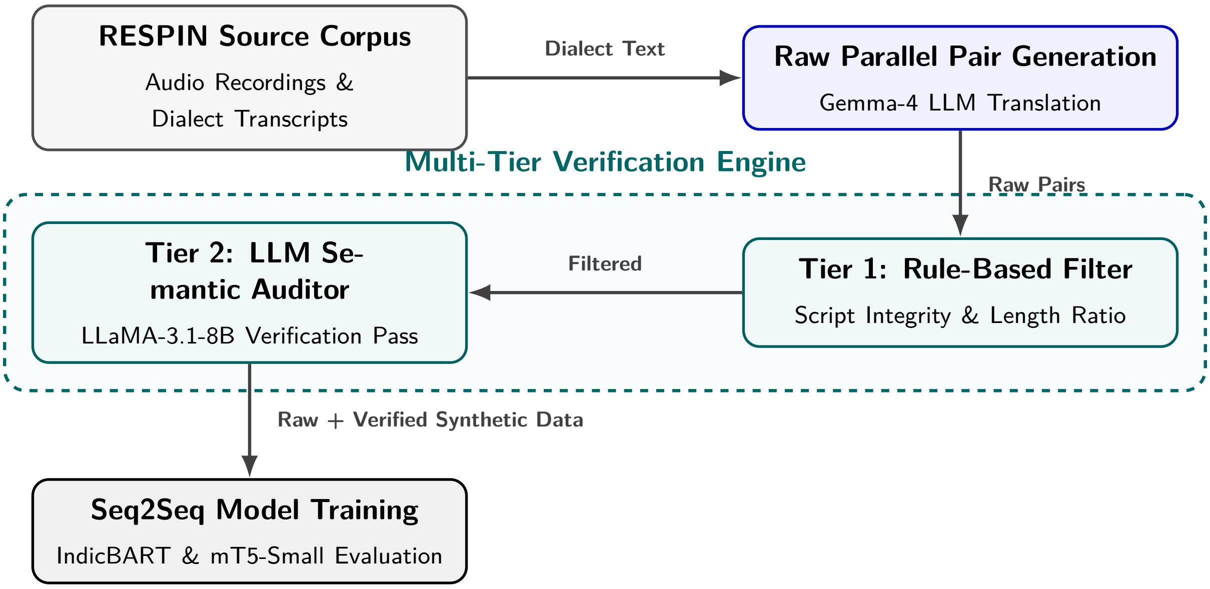

Figure 2: System architecture of the multi-dialect normalization framework.

All generated pairs undergo a three-tier verification and correction pipeline:

1. **Tier 1: Rule-Based Syntactic Filter.** Automated validation checks implement Indic Abugida script normalization principles (Ansary et al., 2024), including:
   - *Devanagari Script Integrity*: Both source and target must possess $> 90\%$ Devanagari Unicode characters within the `U+0900--U+097F` range, automatically discarding unnecessary characters.
   - *Length Ratio Constraint*: $0.5 \leq |L_{\text{dialect}}|/|L_{\text{standard}}| \leq 2.0$ to avoid hallucinations.
   - *Non-Identity Constraint*: $L_{\text{dialect}} \neq L_{\text{standard}}$ to exclude identical string copies.
   - *Minimum Token Count*: Both source and target must contain ≥ 3 whitespace-separated tokens.
2. **Tier 2: LLM Semantic Verification.** A secondary LLM evaluator using `llama-3.1-8b-instant` (accessed via Groq API) performs semantic equivalence verification. The verifier rates on a 1–5 score scale considering semantic preservation, absence of hallucinated entities, absence of Hindi/English contamination and complete standardization of dialects. Pairs scoring < 4.0 are rejected or routed to correction.
3. **Closed-Loop Correction Engine.** Pairs flagged by Tier 1 or Tier 2 enter a two-step correction loop: Step 1 generates corrected translations using specialised prompts based on the pre-defined reason of identified failure.

This leaves us with a total of 16,163 pairs of cleaned translations from the RESPIN Corpus itself.

3.4 Synthetic Data Expansion

After construction the original 16,163 verified parallel pairs through the RESPIN corpus, we applied the same pipeline to synthetically expand it. The expansion phase uses a similar system prompt, inspired from that of translation phase to generate additional pairs from scratch that almost doubles the data size. Out of the 16,993 synthetic parallel pairs generated by the LLM pipeline, our multi-tier verification engine flagged and discarded 821 flawed pairs (an overall corruption/rejection rate of 4.83%), yielding 16,172 clean verified pairs as outlined in Table 2 and Figure 3. Combining with the 16,163 clean original pairs, the final expanded dataset consists of 32,335 verified parallel sentence pairs across all three dialects.

Table 2: Statistics based on generated and verified data

Dialect Name	Original Clean Pairs	Raw Synthetic Generated	Clean Synthetic Verified	Corrupted / Rejected Pairs (%)	Total Clean Expanded Pairs
Southern Konkan	5,576	5,838	5,569	269 (4.61%)	**11,145**
Northern Konkan	5,501	5,825	5,534	291 (5.00%)	**11,035**
Varhadi	5,086	5,330	5,069	261 (4.90%)	**10,155**
Total	**16,163**	**16,993**	**16,172**	**821 (4.83%)**	**32,335**

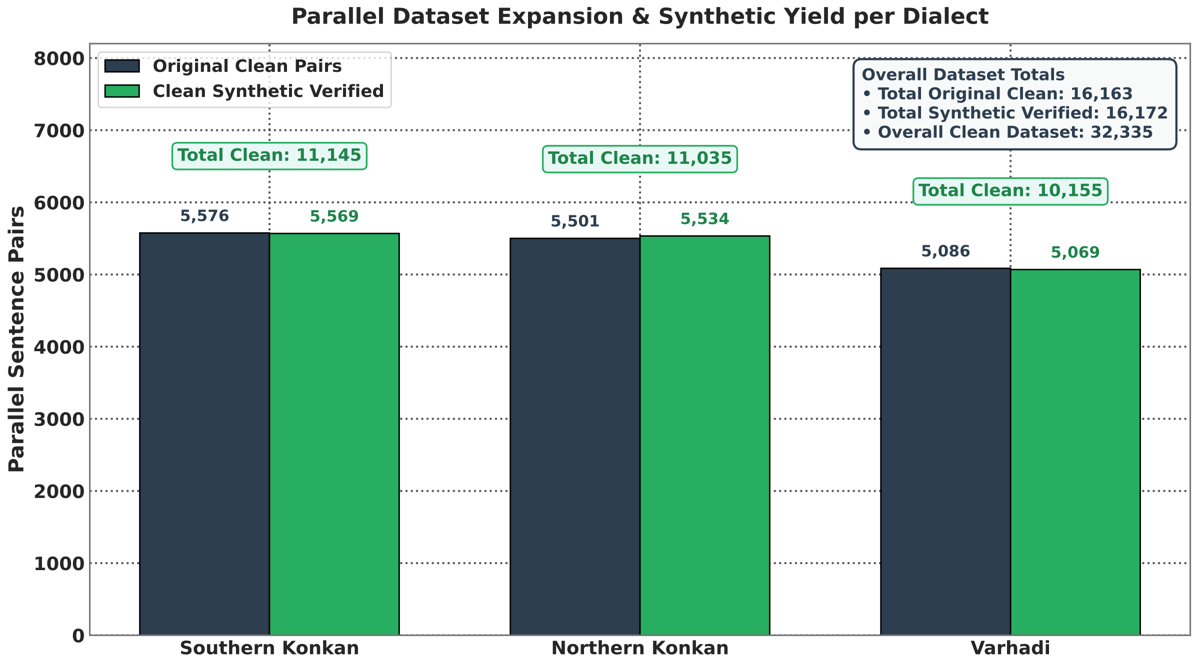

Figure 3: Parallel dataset expansion composition per dialect partition, comparing original parallel pairs against synthetic verified pairs produced by the LLM generation and auditing pipeline.

3.5 Sequence-to-Sequence Model Conditioning and Fine-Tuning Setup

To convert the pre-trained seq2seq language architectures into dedicated dialect normalizers, models are fine-tuned under a 5-fold cross-validation scheme on an 85/15 stratified train-test split. Models were conditioned using specific tags attached before the actual text:

- **Task Normalization Prefixes**: A normalisation tag (`<normalize>`) is attached that instructs the encoder to perform the dialect-to-standard text transformation.
- **Dialect-Source Identification Tags**: Source sentences are prefixed with dialect markers, `<D1>` for Southern Konkan, `<D2>` for Northern Konkan, and `<D4>` for Varhadi for explicitly conditioning the model particularly in the multi-dialect training scenario.
- **Target Standardization Tags**: Decoder target conditioning is done using standard language tags (`<standard_mr>`), enforcing the output alignment into formal Standard Written Marathi.

Table 3 details the underlying architectural properties and hyperparameter configurations for IndicBART (244M) and mT5-small (300M).

Table 3: Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

Property	IndicBART (244M)	mT5-small (300M)
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required `<2mr>` token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)

3.6 Evaluation Protocol

We report model performance across two distinct evaluation settings:

1. **Mean 5-Fold Cross-Validation**: To evaluate model stability and in-domain generalization across the curated parallel datasets, models are trained and evaluated under an 85/15 stratified 5-fold cross-validation scheme (reported in Tables 5–10 and Figures 5–8).
2. **Held-Out RESPIN Test Benchmark**: To measure true zero-shot/unseen generalization, models trained on the complete dataset are evaluated on the official held-out benchmark set (`IISc_RESPIN_test_mr`), consisting of 2,170 unseen utterances across all target sub-dialects (reported in Table 4 and Figure 4).

Results are reported using metrics that are evaluated using SacreBLEU (Post, 2018) and Word Error Rate (WER) scripts:

- **BLEU** (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty, with an exponential smoothing method.
- **chrF++**: Character n-gram F-score combined with word unigrams and bigrams.
- **Normalized WER / CER**: Word Error Rate and Character Error Rate after Devanagari Unicode script standardization.

4 Findings

To ensure transparent analysis, our findings are reported across two distinct evaluation protocols: (1) **Held-Out RESPIN Test Benchmark** (Section 4.1, Table 4, Figure 4): Evaluates true unseen generalization on the official held-out test dataset comprising 2,170 utterances (`IISc_RESPIN_test_mr`) after training models on the complete dataset; and (2) **Mean 5-Fold Cross-Validation** (Sections 4.2–4.4, Tables 5–10, Figures 5–8): Evaluates in-domain model stability and dialectwise synthetic augmentation gains across 5 stratified train/test folds.

4.1 Comprehensive Benchmark on the Held-Out Test Set

Single-dialect training shows that synthetic data augmentation substantially reduces IndicBART performance across all three dialects, while mT5-small remains comparatively stable and shows slight improvements on Northern Konkan and Varhadi. Multi-dialect training produces substantial gains for IndicBART on Southern Konkan (+26.0 BLEU) and Northern Konkan (+6.5 BLEU), while Varhadi decreases slightly (−3.0 BLEU). mT5-small shows smaller but consistently positive gains across all three dialects: +3.0, +0.7, and +1.2 BLEU for Southern Konkan, Northern Konkan, and Varhadi, respectively. The full performance evaluation matrix on the held-out test set is detailed in Table 4 and Figure 4.

Table 4: Comprehensive Evaluation Matrix on `IISc_RESPIN_test_mr` Benchmark Set (2,170 Utterances)

Model Scope	Training Data	Dialect / Subset	Utts	IndicBART (244M)		mT5-Small (300M)	
				BLEU (↑)	WER (↓)	BLEU (↑)	WER (↓)
Single-Dialect	Original	Southern Konkan	559	**57.80**	**26.60%**	43.86	34.37%
		Northern Konkan	540	**90.13**	**6.46%**	79.46	11.95%
		Varhadi	516	**83.59**	**10.19%**	74.81	14.73%
	Synthetically Expanded	Southern Konkan	559	24.06	60.32%	**44.36**	**32.75%**
		Northern Konkan	540	65.41	28.70%	**79.76**	**12.61%**
		Varhadi	516	67.97	25.43%	**76.90**	**14.19%**
Multi-Dialect	Original	Southern Konkan	559	26.08	52.98%	**40.94**	**35.34%**
		Northern Konkan	540	47.92	32.64%	**77.50**	**14.65%**
		Standard Benchmark	555	96.12	3.12%	**96.65**	**1.91%**
		Varhadi	516	73.04	26.96%	**73.47**	**16.16%**
		Overall Benchmark	**2,170**	62.98	30.13%	**72.18**	**17.23%**
	Synthetically Expanded	Southern Konkan	559	**52.15**	36.50%	43.88	**34.96%**
		Northern Konkan	540	54.35	25.65%	**78.16**	**13.55%**
		Standard Benchmark	555	96.80	1.65%	**97.39**	**1.37%**
		Varhadi	516	69.96	21.04%	**74.74**	**15.57%**
		Overall Benchmark	**2,170**	**76.50**	**16.23%**	73.48	16.58%

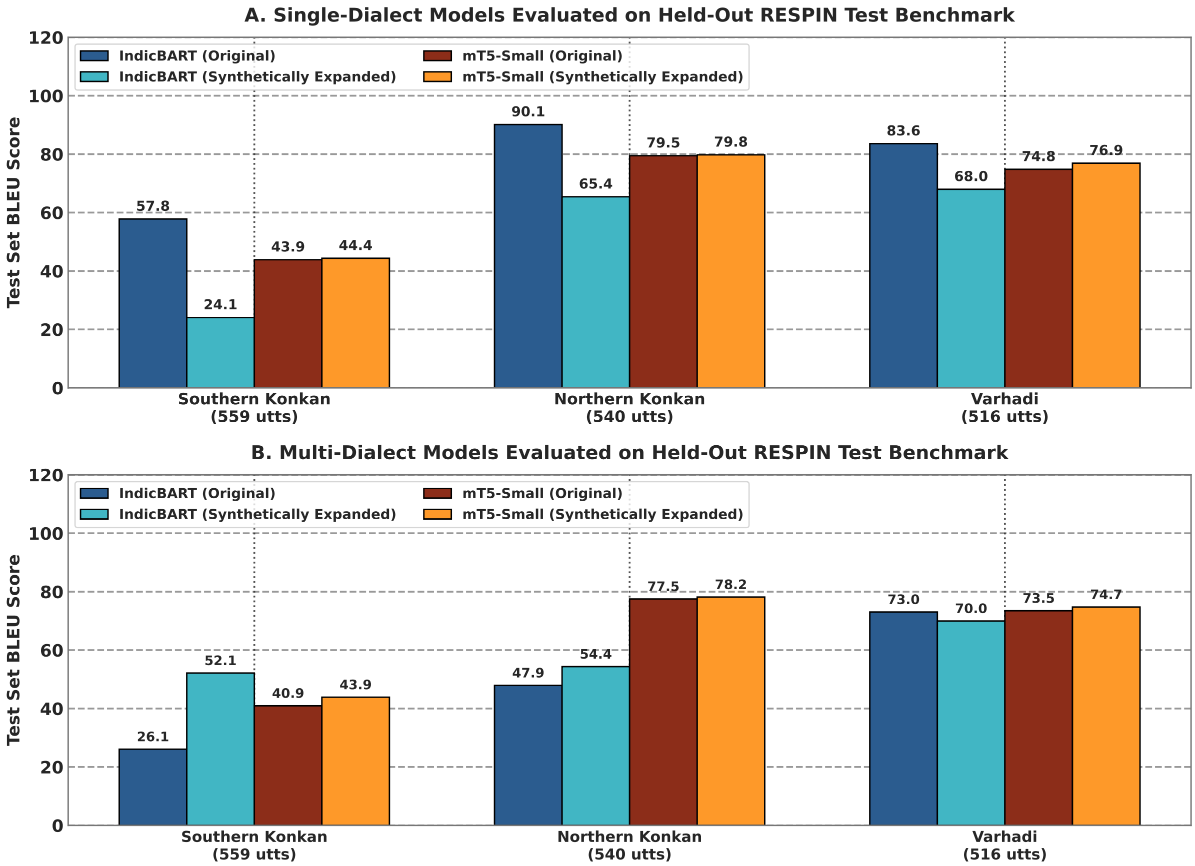

Figure 4: Comprehensive sequence-to-sequence evaluation matrix on the held-out RESPIN test benchmark (2,170 utterances), comparing IndicBART (244M) and mT5-small (300M) across single-dialect and multi-dialect training configurations.

4.2 Single-Dialect vs. Multi-Dialect 5-Fold Cross-Validation Results

Table 5 presents the performance matrix on original single-dialect benchmarks, and Table 6 explains the dialect-wise breakdown of the multi-dialect model on original data. Table 7 presents the performance matrix on synthetically expanded single-dialect benchmarks, and Table 8 details the dialectwise evaluation breakdown of the multi-dialect model on expanded data. Comparative 5-fold cross-validation performance shifts and multi-dialect matrices are illustrated in Figure 5 and Figure 6.

Table 5: Benchmark Performance Matrix on Original Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	5,576	837 / 836	**48.54**	**78.35**	46.31	74.43
Northern Konkan (D2)	5,501	826 / 825	**60.58**	**80.30**	60.31	77.34
Varhadi (D4)	5,086	763 / 762	76.63	90.93	**81.00**	**91.21**

Table 6: Benchmark Performance Matrix on Original Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	16,163	837 / 836	26.21	53.15	**48.28**	**75.04**
Northern Konkan (D2)	16,163	826 / 798	47.59	67.16	**62.35**	**78.92**
Varhadi (D4)	16,163	763 / 790	72.75	85.63	**79.57**	**90.51**
Overall Combined	**16,163**	**2,426 / 2,424**	**48.17**	**68.58**	**63.29**	**81.37**

Table 7: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	11,145	1,672 / 1,671	47.25	64.69	**65.10**	**82.88**
Northern Konkan (D2)	11,035	1,656 / 1,655	40.51	62.21	**62.07**	**79.29**
Varhadi (D4)	10,155	1,524 / 1,523	73.62	86.75	**78.89**	**90.59**

Table 8: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	32,335	1,672 / 1,710	52.06	72.15	**66.62**	**83.57**
Northern Konkan (D2)	32,335	1,656 / 1,627	49.08	69.90	**62.72**	**79.63**
Varhadi (D4)	32,335	1,524 / 1,513	70.27	84.27	**78.73**	**90.46**
Overall Combined	**32,335**	**4,852 / 4,850**	**57.12**	**75.49**	**69.51**	**84.58**

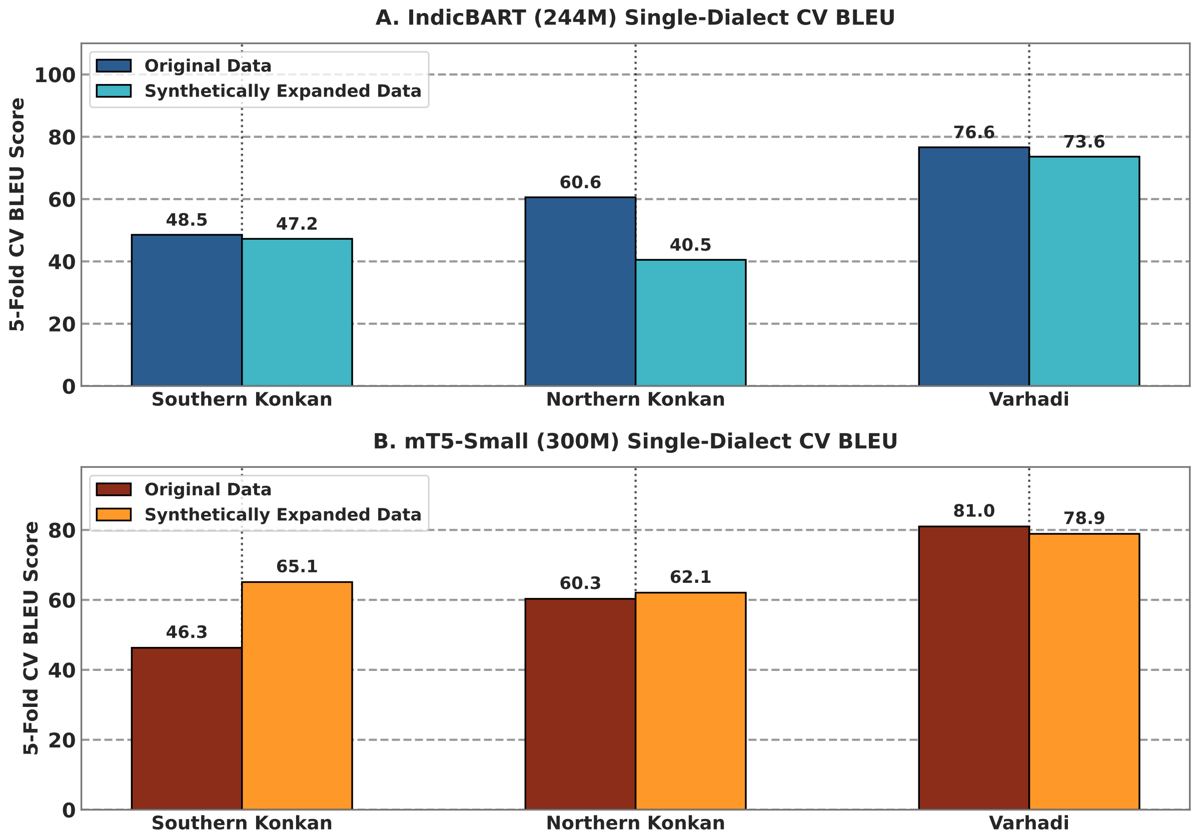

Figure 5: Single-dialect 5-fold cross-validation performance shift comparing Original and Synthetically Expanded models for IndicBART (244M) and mT5-small (300M).

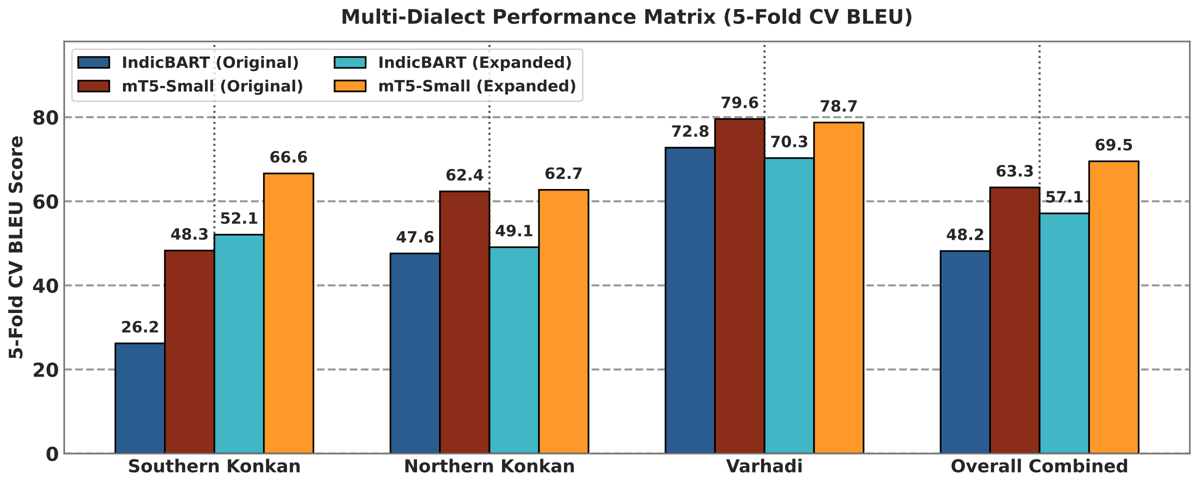

Figure 6: Multi-dialect 5-fold cross-validation benchmark comparison across model architectures and corpora partitions.

4.3 Impact of Synthetic Data Augmentation

Table 9 shows the effect of dataset scaling by comparing performance across original and synthetically expanded corpora for each model architecture. Overall, multi-dialect models exhibit substantial gains from synthetic data augmentation for most dialects. IndicBART shows particularly large improvements for Southern Konkan (+26.0 BLEU) and Northern Konkan (+6.5 BLEU), although its Varhadi performance decreases by 3.0 BLEU. mT5-small shows smaller but consistently positive gains across all three dialects. Visual representation of net BLEU score changes is provided in Figure 7.

Table 9: Impact of Synthetic Data Augmentation: Original vs. Synthetically Expanded Data BLEU Change (5-Fold CV)

Evaluated Subset	IndicBART BLEU			mT5-Small BLEU		
	Original	Expanded	Δ	Original	Expanded	Δ
Southern Konkan (D1)	26.21	52.06	**+25.85**	48.28	**66.62**	**+18.34**
Northern Konkan (D2)	47.59	49.08	**+1.49**	62.35	**62.72**	**+0.37**
Varhadi (D4)	**72.75**	70.27	−2.48	**79.57**	78.73	−0.84
Overall Combined	**48.17**	**57.12**	**+8.95**	**63.29**	**69.51**	**+6.22**

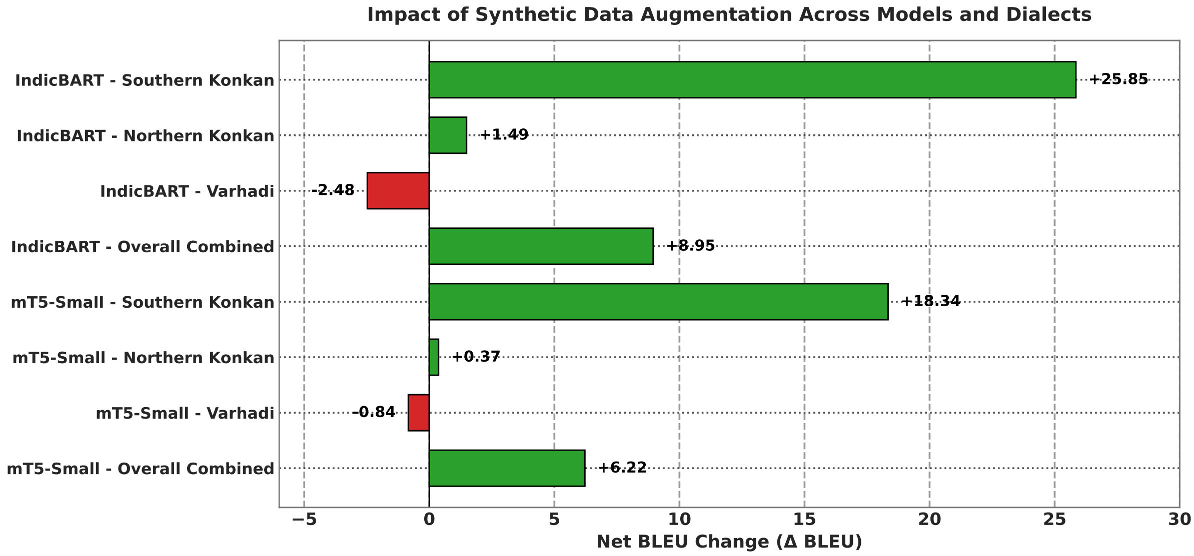

Figure 7: Impact of LLM synthetic data expansion (Δ BLEU score comparing synthetically expanded vs. original 5-fold cross-validation benchmarks).

Metrics from Table 9 hint a relationship with the performance of models with their architectures and pre-training practices. On the combined multi-dialect benchmarks, mT5-small outperforms IndicBART across both the original and expanded datasets (63.29 vs. 48.17 BLEU and 69.51 vs. 57.12 BLEU, respectively). This could hint us towards a better subword vocabulary capacity of mT5-small; 250,000-token vocabulary segments show lower fragmentation than IndicBART's 64,000 Devanagari unified vocabulary. Also, mT5's relative positional encoding handles merger of dialectal suffix better than IndicBART's absolute sinusoidal encodings.

4.4 Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

We analysed the performance of our verification and filtering engine by conducting a detailed comparative analysis with models trained on the Unverified dataset. It had around 19,914 pairs that included the 3,751 rejected pairs that contained contaminated translations. Table 10 and Figure 8 demonstrate the BLEU and chrF++ score recovery provided by multi-tier verification.

Table 10: Impact of Multi-Tier Verification Engine: Raw Unverified vs. Filtered Clean Synthetic Data

Model	Pipeline State	Total Pairs	BLEU	chrF++	Δ BLEU	Δ chrF++
IndicBART (244M)	Raw Unverified Data	19,914	43.09	64.54	−14.03	−10.95
	Filtered Clean Data	32,335	**57.12**	**75.49**	**+14.03**	**+10.95**
mT5-Small (300M)	Raw Unverified Data	19,914	61.21	80.18	−8.30	−4.40
	Filtered Clean Data	32,335	**69.51**	**84.58**	**+8.30**	**+4.40**

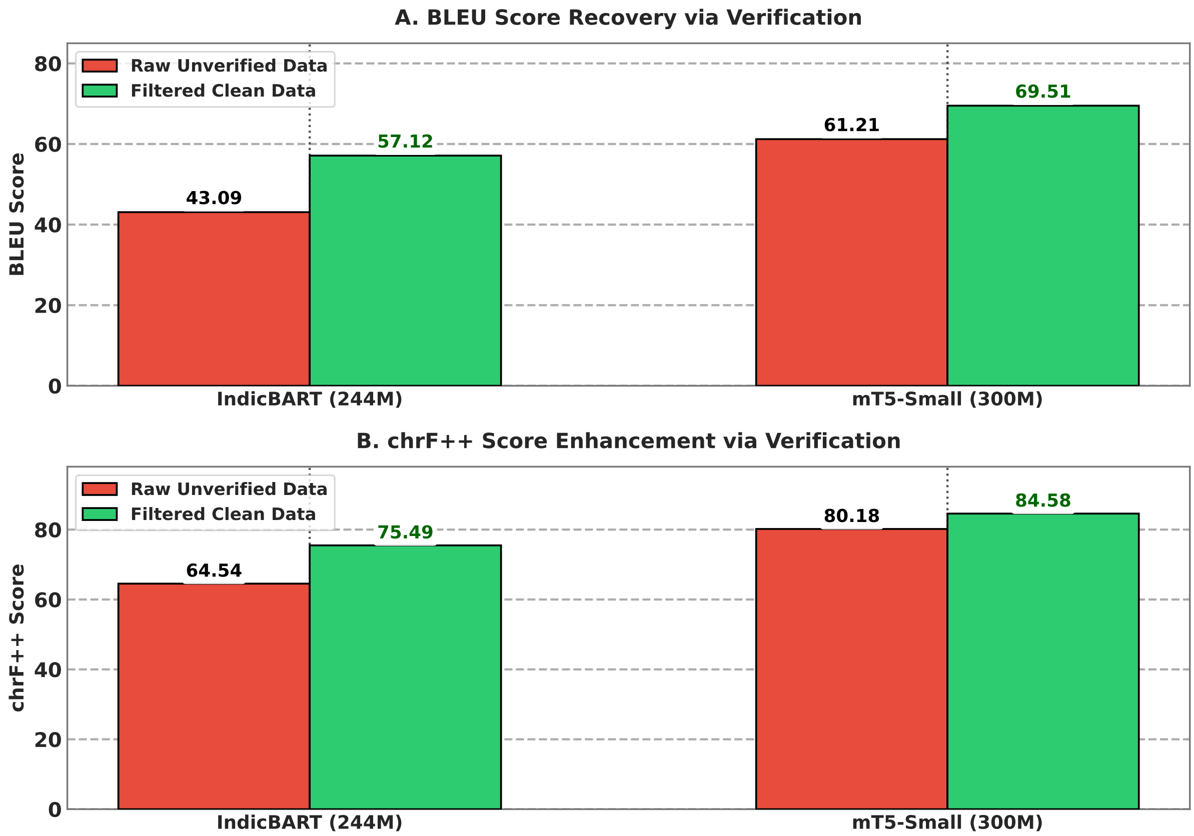

Figure 8: Multi-tier verification engine ablation: (A) BLEU score recovery and (B) chrF++ score enhancement comparing models trained on raw unverified synthetic outputs vs. filtered clean synthetic data.

Training on raw unverified synthetic data causes severe performance degradation (−14.03 BLEU for IndicBART, −8.30 BLEU for mT5-small). Unfiltered LLM outputs had hallucinated sentences which corrupted the gradients during backpropagation. Multi-tier verification (rule-based script filtering and LLM semantic auditing) removes 3,751 flawed pairs successfully recovering BLEU scores to 57.12 (IndicBART) and 69.51 (mT5-small).

IndicBART shows sensitivity to noise, suffering a 24.6% relative BLEU drop on unverified data (57.12 $\rightarrow$ 43.09). In contrast, mT5-small demonstrates robustness with a smaller relative drop of 11.9% (69.51 $\rightarrow$ 61.21). This could be directly attributed to mT5's larger 300M parameter capacity and extensive pre-training corpus providing stronger immunity against the noise. This highlights the importance of the quality audit of data while using LLMs for generating or correcting the data.

5 Conclusion

This study presents a scalable, reproducible framework for multi-dialect Marathi text normalization, addressing the critical scarcity of parallel corpora for low-resource Indic varieties. By replacing expensive manual annotations with a synthetic parallel generation pipeline combining Gemma-4 translation with a multi-tier verification engine (rule-based script filtering and LLaMA-3.1-8B semantic auditing), we successfully expanded the clean parallel corpus from 16,163 to 32,335 verified pairs across Southern Konkan, Northern Konkan, and Varhadi dialects. Empirical evaluations demonstrate that mT5-small achieves 69.51 BLEU and 84.58 chrF++ on the expanded multi-dialect benchmark, outperforming IndicBART by 12.39 BLEU. Crucially, multi-tier verification prevented synthetic quality collapse (+14.03 BLEU recovery over unverified data), while dataset expansion mitigated cross-dialect parameter competition and maintained near-perfect Standard Marathi normalization (97.39 BLEU, 1.37% WER) on held-out RESPIN benchmarks. While these findings establish a strong blueprint for Indic sub-dialect normalization, minor constraints remain: LLM-driven verification pipelines introduce computational and latency overheads, synthetic expansion can occasionally yield slight precision dilution on already high-performing dialects like Varhadi, and intrinsic text metrics should be complemented by downstream task evaluations (e.g., ASR post-processing) in future work. Overall, our verified synthetic pipeline provides a scalable foundation for expanding digital linguistic inclusion across low-resource regional dialects.

References

Abdul-Mageed, M., Zhang, C., Elmadany, A., and Nagoudi, E. M. (2023). Nadi 2023: The fourth nuanced arabic dialect identification and translation shared task. In *Proceedings of the Arabic NLP Conference (ArabicNLP @ EMNLP 2023)*, pages 512–528.

Abe, K., Inui, K., and Matsumoto, Y. (2018). Multi-dialect neural machine translation for japanese dialects. In *Proceedings of the 27th International Conference on Computational Linguistics (COLING 2018)*, pages 2105–2116.

Ansary, N., Adib, Q. A. R., Reasat, T., Sushmit, A. S., Humayun, A. I., Mehnaz, S., Fatema, K., Or Rashid, M. M., and Sadeque, F. (2024). Unicode normalization and grapheme parsing of indic languages. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 17019–17030. ELRA Language Resource Association.

Blaschke, V., Purschke, C., Schütze, H., and Plank, B. (2024). What do dialect speakers want? a survey of attitudes towards language technology for german dialects. *arXiv preprint arXiv:2402.11968*.

Bouamor, H., Habash, N., Salameh, M., et al. (2018). The madar parallel corpus of arabic dialects. In *Proceedings of the 11th Language Resources and Evaluation Conference (LREC 2018)*, pages 3375–3381.

Dabre, R., Shrotriya, H., Kunchukuttan, A., Puduppully, R., Khapra, M. M., and Kumar, P. (2022). Indicbart: A pre-trained model for indic natural language generation. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 1849–1863.

de Gibert, O. et al. (2024). Scaling low-resource mt via synthetic data generation with llms. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.

Dhasmana, A., Srivastava, A., and Chiang, D. (2026). Dialect matters: Cross-lingual asr transfer for low-resource indic language varieties. In *Proceedings of the Thirteenth Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ ACL 2026)*, pages 145–156. Association for Computational Linguistics.

Dimakis, A., Pavlopoulos, J., and Anastasopoulos, A. (2025). Dialect normalization using large language models and morphological rules. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 23696–23714. Association for Computational Linguistics.

Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Yang, A., Fan, A., and et al. (2024). The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Eryani, F., Habash, N., Al-Shargi, F., et al. (2020). A normalization corpus for five arabic dialects. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC 2020)*, pages 5750–5757.

Gaikwad, G., Panchal, D., Deshmukh, O., and Patil, S. (2025). Marathi dialect to pure marathi using rag & ai. *International Research Journal of Engineering and Technology (IRJET)*, 12(11):787–791.

Google DeepMind Gemma Team (2024). Gemma 4: Open language models. *arXiv preprint arXiv:2408.00118*.

Habash, N., Diab, M., and Rambow, O. (2012). Conventional orthography for dialectal arabic. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC '12)*, pages 711–718.

Hämäläinen, M., Partanen, N., and Alnajjar, K. (2020). Normalizing swedish dialects spoken in finland. In *Proceedings of the 7th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2020)*, pages 55–62.

Hämäläinen, M., Partanen, N., and Rueter, J. (2022). Generating dialectal finnish for dialect normalization of estonian dialects. In *Proceedings of the 9th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2022)*, pages 71–79.

Hsieh, C.-Y. et al. (2023). Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. *arXiv preprint arXiv:2305.02301*.

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., and Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.

Joshi, A., Dabre, R., Kanojia, D., Li, Z., Zhan, H., Haffari, G., and Dippold, D. (2025). Natural language processing for dialects of a language: A survey. *ACM Computing Surveys*, 57(6):1–37.

Joshi, P., Santy, S., Budhiraja, A., Bali, K., and Choudhury, M. (2020). The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 6282–6293.

Joshi, R., Goel, P., and Joshi, R. (2022). L3cube-mahanlp: Marathi natural language processing datasets, models, and library. In *Proceedings of the WILDRE-6 Workshop, LREC 2022*, pages 10–17.

Kakwani, D., Kunchukuttan, A., Golla, S., Gokul, N. C., Bhattacharyya, P., Khapra, M. M., and Kumar, P. (2020). Indicnlpsuite: Monolingual corpora, evaluation benchmarks and pre-trained multilingual language models for indic languages. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4948–4961. Association for Computational Linguistics.

Koenecke, A., Nam, A., Lake, E., Nudell, J., Quartey, M., Mengesha, Z., Toups, C., Rickford, J. R., Jurafsky, D., and Goel, S. (2020). Racial disparities in automated speech recognition. *Proceedings of the National Academy of Sciences*, 117(14):7684–7689.

Kresic, M. and Abbas, N. (2024). Normalizing swiss german dialects with the power of large language models. *Procedia Computer Science*, 244:287–295.

Kulkarni-Joshi, S. (2023). Variation and change in dialects of marathi: A social-dialectological approach. In Chandra, P., editor, *Variation in South Asian Languages: From Macro to Micro-Differences*, pages 207–236. Springer.

Kumar, A., Shrivastava, M., and Bhattacharyya, P. (2024). Indic-dialect: Benchmark dataset for indic dialect classification and translation. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 15201–15212. ELRA Language Resource Association.

Kumar, S., Singh, A., G, D., Amartyaveer, Bandekar, J., Murthy, S., Sharma, S., Badiger, S., Udupa, S., Nagireddi, A., Raghavan, S. K. M., Saxena, R., Nanavati, J., Nanavati, R., Sridharan, J., Mehta, A., Khuraishi, A. K. S., Mora, S. P. R., Venkataramakrishnan, P., Date, G., P, K., and Ghosh, P. K. (2025). RESPIN-S1.0: A read speech corpus of 10000+ hours in dialects of nine indian languages. In *Advances in Neural Information Processing Systems (NeurIPS 2025) Track on Datasets and Benchmarks*, pages 1–13.

Kuparinen, O., Miletić, A., and Scherrer, Y. (2023). Dialect-to-standard normalization: A large-scale multilingual evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 13814–13828. Association for Computational Linguistics.

Lusetti, M., Ruzsics, T., Samardžić, T., and Scherrer, Y. (2018). Impact of target language models on swiss german dialect normalization. In *Proceedings of the 5th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2018)*, pages 133–141.

Mukherjee, S., Mitra, A., Jawahar, G., Agarwal, S., Palangi, H., and Awadallah, A. (2023). Orca: Progressive learning from complex explanation traces of gpt-4.

Mulla, R. and Kumar, B. S. (2022). Text-independent automatic dialect recognition of marathi language using spectro-temporal characteristics of voice. *International Journal on Recent and Innovation Trends in Computing and Communication (IJRITCC)*, 10(2s):313–321.

Partanen, N., Hämäläinen, M., and Alnajjar, K. (2019). Dialect text normalization to normative standard finnish. In *Proceedings of the 5th Workshop on Noisy User-generated Text (W-NUT @ EMNLP 2019)*, pages 141–146. Association for Computational Linguistics.

Pawar, P. S., Saini, J. R., and Vaidya, S. (2026). Automatic marathi dialect classification in noisy and non-noisy environments. *Procedia Computer Science*, 282:288–303.

Pecher, B. et al. (2024). Better as generators than classifiers: Leveraging llms and synthetic data for low-resource multilingual classification. In *Findings of the Association for Computational Linguistics: EACL 2024*.

Post, M. (2018). A call for clarity in reporting bleu scores. In *Proceedings of WMT 2018*, pages 186–191.

Pratap, V., Saraf, A., and Kumar, P. (2024). Wfst-guided neural text normalization for indic languages. In *Proceedings of Interspeech 2024*, pages 3120–3124.

Scherrer, Y. and Ljubešić, N. (2016). Normalising swiss german dialect text using character-level machine translation. In *Proceedings of the 3rd Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2016)*, pages 117–126.

Taori, R., Gulrajani, I., Zhang, T., Dubois, Y., Li, X., Guestrin, C., Liang, P., and Hashimoto, T. B. (2023). Alpaca: A strong, replicable instruction-following model. *Stanford Center for Research on Foundation Models*.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS 2017)*, pages 5998–6008.

Vyavhare, V. A., Divekar, S. N., Pawar, H., Kale, K., and Jagtap, S. (2026). Marathi dialect translator. *International Journal of Scientific Research in Science, Engineering and Technology (IJSRSET)*, 13(3):254–261.

Wang, Y., Kordi, Y., Mishra, S., Liu, A., Smith, N. A., Khashabi, D., and Hajishirzi, H. (2022). Self-instruct: Aligning language models with self-generated instructions.

Xue, L., Constant, N., Roberts, A., Kale, M., Al-Rfou, R., Siddhant, A., Barua, A., and Raffel, C. (2021). mt5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of NAACL-HLT 2021*, pages 483–498.

Zampieri, M., Nakov, P., and Scherrer, Y. (2020). Natural language processing for similar languages, varieties, and dialects: A survey. *Natural Language Engineering*, 26(6):605–637.

Zhao, Y. and Chodroff, E. (2022). Acoustic-phonetic variation in mandarin dialect corpora. *Journal of Phonetics*, 94:101172.